Antonio Colón

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EDUCATION

University of Michigan **Ann Arbor, MI**
Bachelor of Science in Computer Science, Minor in Mathematics *May 2026*
Coursework: Computer Architecture, Security, Web Systems, Databases, Probability, Mathematics of Finance
Extracurriculars: Team Captain – Club and Varsity Soccer, Michigan Student AI Lab, Michigan Rock Climbing Club

WORK EXPERIENCE

International Finance Bank (IFB) **Miami, FL**
Cybersecurity Intern *Jul. 2024 – Aug. 2024*

- Ran phishing simulation campaigns in KnowBe4 and led weekly presentations of results and security awareness training to employees, reducing the bank's phish-prone percentage by ~50% over one summer.
- Pioneered the integration of Business Impact Analysis (BIA) data into a new SaaS platform, streamlining data storage and accessibility for the bank.
- Supported the trademark registration of the bank's new logo, helping protect the brand against phishing, impersonation, and fraudulent accounts.

International Finance Bank (IFB) **Miami, FL**
Software Engineering Intern *Jul. 2023 – Aug. 2023*

- Identified performance bottlenecks in client reporting queries, removing redundant joins and consolidating repeated logic across several of them to reduce report generation time by ~40%.
- Accomplished the creation of a company-wide Power BI dashboard, improving data reporting across the organization, and further improving relationships with clients.
- Invited back for a second summer internship in cybersecurity based on performance and supervisor feedback.

PROJECTS

CartWise *Jan. 2026 – Present*
Founder

- Founded CartWise (trycartwise.com), a full-stack web app that turns recipes into one-click shopping carts by structuring ingredients and mapping them to Walmart SKUs; integrated Walmart APIs for automated cart creation, deployed on AWS with DynamoDB for data storage.

MODUS AI *Feb. 2026*

- Built Modus AI, a retrieval-augmented tutoring system for university courses that answers questions from lecture transcripts with citations and timestamp links, using retrieval and filtering to reduce hallucinated answers.
- Deployed Modus AI in a live university course and placed 5th of 100+ teams at the Oracle x University of Michigan hackathon.

Scalable Search Engine *Apr. 2025*

- Built a multi-service search engine with a multi-stage MapReduce pipeline that cleaned, indexed, and partitioned crawled documents into segmented inverted indexes.
- Combined TF-IDF and PageRank signals for relevance ranking; served concurrent queries across parallel Flask index servers and merged results with document metadata.

Distributed MapReduce *Mar. 2025*

- Built a fault-tolerant MapReduce framework with a central manager that scheduled map and reduce tasks and tracked worker state via TCP messaging and UDP heartbeats.
- Detected and reassigned failed worker tasks; managed partitioned intermediate output, external sorting, and merging of results across networked worker processes.

SKILLS

Languages: C++, Python, C, Java, JS, SQL, HTML
Frameworks & Cloud: React, Flask, Docker, AWS, DynamoDB, Render, Cloudflare R2, REST APIs
Tools: Git, Claude Code, Antigravity